

# Percentiles & Quartiles

## Tutorial - 6

### 1. Percentage vs. Percentile

Hello students! Today we are learning about a very important topic: **Percentiles and Quartiles**. Before we jump into percentiles, we must understand how it is completely different from a normal "Percentage".

**Percentage:** A simple math calculation showing a part out of 100. It focuses only on your own score or count.

#### Real-Life Example:

**Percentage Example:** Let's take a list of numbers: 1, 2, 3, 4, 5.

*Question:* What is the percentage of Even numbers in this list?

*Answer:* The even numbers are 2 and 4 (Total = 2 numbers). The total numbers in the list = 5.

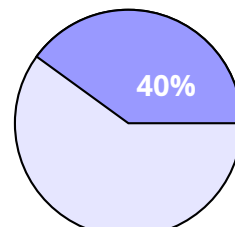
Percentage =  $\frac{2}{5} \times 100 = 40\%$ .

So, 40% of the numbers in the list are even!

#### Practice Question:

**Q:** If you score 80 marks out of 100 in a math test, what is your percentage?

**A:** Your percentage is 80%. It only cares about your marks, not the rest of the class.



**Percentage:**  
Focuses on "Part  
out of Total"

### 2. What is a Percentile?

**Definition:** A Percentile is a value below which a certain percentage of observations lie. It shows your **Ranking** compared to everyone else!

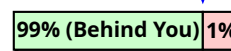
If you have heard of big entrance exams like CAT, GATE, or SAT, they don't give you normal marks; they give you a **Percentile Score**. A percentile does not tell you how many questions you got right. It tells you how many people you defeated!

#### Real-Life Example:

##### The 99th Percentile:

Imagine you get your test results, and it says you scored in the **99th Percentile**. What does this mean? It means your score is **better than 99% of all the students** who took the exam! If 100 students wrote the test, you scored higher than 99 of them. You are at the very top!

**YOU!**  
(99th %ile)



## Topic Summary: Percentage vs. Percentile

**Percentage** = Your own individual score (e.g., 80 out of 100).

**Percentile** = Your ranking compared to others (e.g., scoring better than 80% of the class).

### 3. Finding the Percentile Ranking

Let's say we have a dataset and we want to find the percentile ranking of a specific number.

**Formula:**

$$\text{Percentile Rank of } X = \left( \frac{\text{Number of values below } X}{\text{Total number of values}} \right) \times 100$$

**Real-Life Example:**

**Our Dataset (Sorted):**

2, 2, 3, 4, 5, 5, 5, 6, 7, 8, 8, 8, 8, 8, 9, 9, **10**, 11, 11, 12.

Total numbers ( $n$ ): 20.

**Question:** What is the percentile ranking of the number **10**?

- **Step 1:** Count how many numbers are strictly *smaller* than 10. (If you count them, there are exactly 16 numbers before 10).
- **Step 2:** Use the formula:  $\left(\frac{16}{20}\right) \times 100$
- **Step 3:** Calculate!  $\frac{16}{20} = 0.8$ . And  $0.8 \times 100 = 80$ .

**Result:** The value 10 is at the **80th Percentile**. This means the number 10 is greater than 80% of the entire distribution!

**Practice Question:**

**Q (Assignment):** In the same list above, what is the percentile ranking of the number **11**?

**A:** There are 17 numbers below 11. So,  $\left(\frac{17}{20}\right) \times 100 = 85$ th Percentile.

Finding Rank  
of 10

|                             |    |  |
|-----------------------------|----|--|
| 16 Numbers<br>(Smaller than | 10 |  |
|-----------------------------|----|--|

↓

$$\frac{16}{20} \times 100$$

= **80th %ile**

## 4. Finding the Value at a Percentile

What if we do the reverse? If I ask you, "What value exists exactly at the 25th Percentile?", how do we find it?

**Formula to find the Index position:**

$$\text{Value Index} = \left( \frac{\text{Percentile}}{100} \right) \times (n + 1)$$

### Real-Life Example:

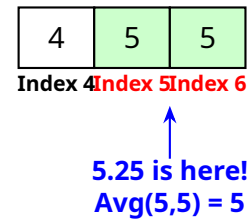
Using the same dataset of 20 numbers...

**Question:** What value is at the **25th Percentile**?

- **Step 1:** Find index =  $\left( \frac{25}{100} \right) \times (20 + 1)$
- **Step 2:**  $0.25 \times 21 = 5.25$ .
- **Step 3:** The index is 5.25. Since there is no "5.25th" position, we look at the 5th and 6th numbers in our sorted list and take their average.
- **Step 4:** In our list, the 5th number is **5**, and the 6th number is **5**. Average =  $\frac{5+5}{2} = 5$ .

**Result:** The value 5 sits at the 25th Percentile!

### Finding Index 5.25



## 5. What are Quartiles?

**Definition:** Quartiles are special percentiles that divide our data perfectly into four quarters (parts).

Instead of saying random percentiles, we give special names to specific milestones:

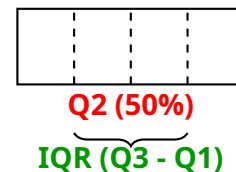
- **25th Percentile** is called the **First Quartile (Q1)**.
- **50th Percentile** is called the **Second Quartile (Q2)** (This is also the Median!).
- **75th Percentile** is called the **Third Quartile (Q3)**.

### Important Note & Extra Info:

#### Interquartile Range (IQR):

If you subtract Q1 from Q3 (**IQR = Q3 - Q1**), you get a magical number! In the next video, we will learn how to use this IQR number to find **Outliers** (weird data points) using a "Box Plot".

Q1 (25%) Q3 (75%)



## Overall Revision Summary

- **Percentage vs. Percentile:** Percentage is your raw score out of 100. Percentile is your *rank* compared to everyone else. (90th percentile = better than 90% of people).
- **Percentile Rank Formula:**  $(\text{Count of numbers smaller than } X \div \text{Total count}) \times 100$ .
- **Value at Percentile Formula:**  $\text{Index} = (\text{Percentile} \div 100) \times (n + 1)$ . If the index has a decimal (like 5.25), average the two numbers it sits between!
- **Quartiles:**
  - Q1 = 25th Percentile
  - Q3 = 75th Percentile
  - IQR (Interquartile Range) =  $Q3 - Q1$ . This formula is the secret key to finding Outliers!

*Keep practicing! In the next lesson, we will use Quartiles to master Box Plots and the 5-Number Summary!*